Homework 3

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```
# read in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(patchwork)
library(ggeffects)
```

```r
# read in data and create new object
kelp <- read_csv(
  here("data", "temp-kelp.csv"))

# creating new data frame with personal data
personal_data <- read_csv(here("data", "personal-datasheet.csv")) |>
  # standardizing column names
  clean_names() |>
  # dropping rows with na observations
  drop_na() |>
  # changing the stop column to display TRUE/FALSE instead of 1/0
  mutate(stop = as.logical(stop))

# creating new data frame for model predictions
personal_num <- personal_data |>
  # changing depature time column to numeric value
  mutate(departure_time = as.numeric(departure_time)) |>
  # selecting needed columns for model
  select(departure_time, walk_minutes)
```

Problem 1. Giant kelp fronds

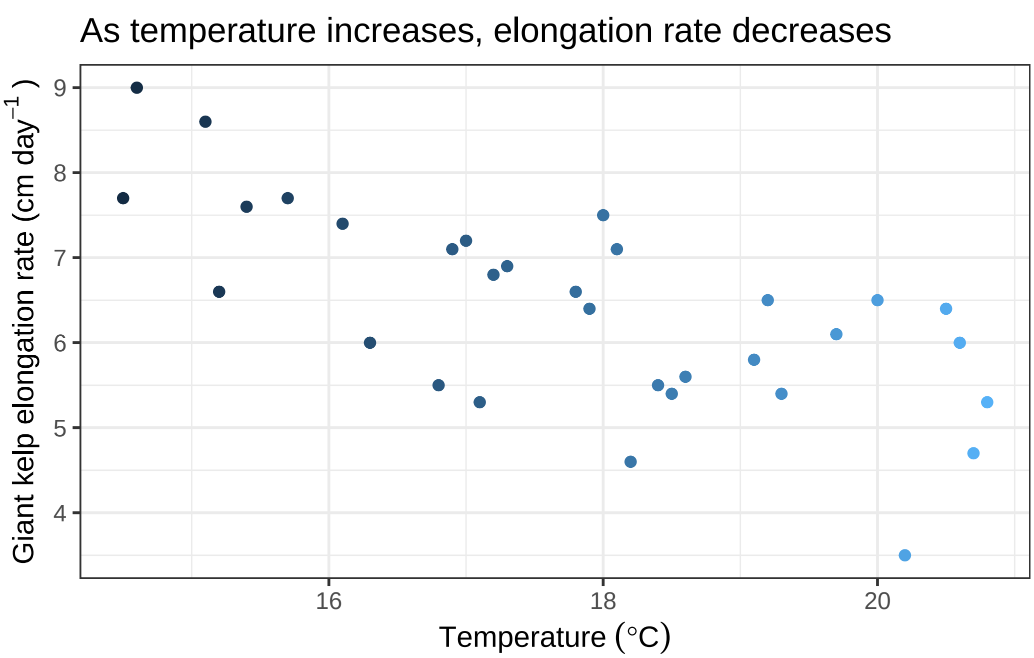
a.

There are two appropriate tests for testing the strength of the relationship between temperature and giant kelp frond elongation rate (correlation). These two are the Pearson's correlation, which assumes normally distributed continuous data, and Spearman's rank correlation, which does not require normality but does require continuous data. Pearsons'scorrelation is the correct test to choose when there are no outliers in the data and the distribution is clearly normal, in all other cases Spearman's is more appropriate.

b.

```r
# base layer: ggplot, data from kelp data frame
ggplot(data = kelp,
       mapping = aes(
         x = temp_c, # x-axis: temperature
         y = kelp_elong, # y-axis: elongation rate
```

```
       color = temp_c)) + # color gradient by temperature
  # first layer: scatter plot
  geom_point() +
  # labeling axes and adding title
  labs(x = expression(Temperature~(degree*C)),
       y = expression("Giant kelp elongation rate (cm day"^-1~")"),
       title = "As temperature increases, elongation rate decreases") +
  # changing theme
  theme_bw() +
  # removing legend
  theme(legend.position = "none")
```

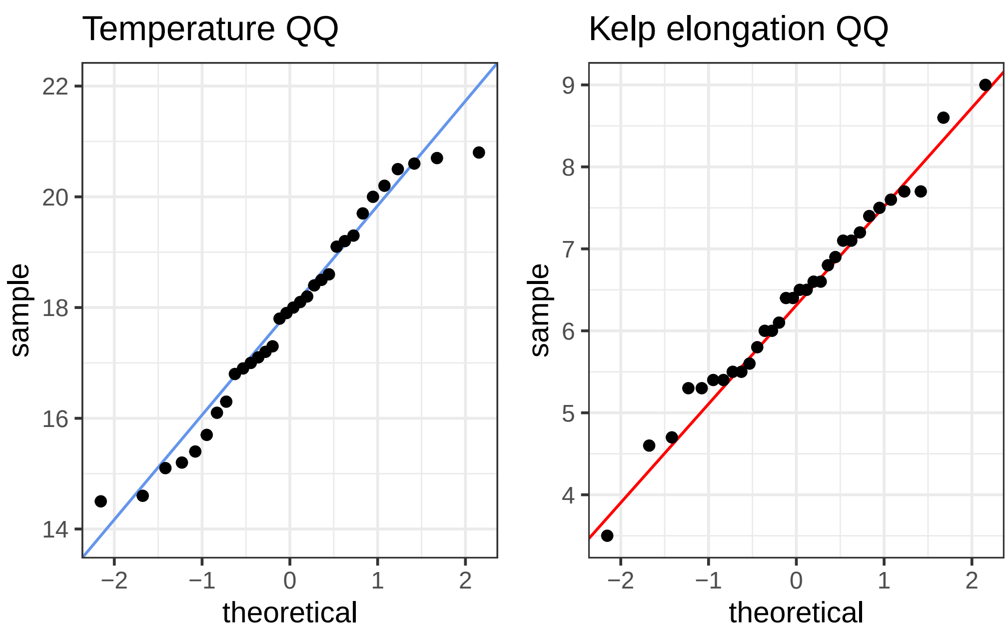


c.

Assumption check

Relationship appears linear, likely meaning Pearson's is the accurate test to run. Going to check normality of each variable, as each is continuous, to confirm that Pearson's is the correct test.

```
# creating object for temperature QQ plot
tempQQ <-
# baselayer: ggplot
ggplot(data = kelp, # using kelp data frame
       aes(sample = temp_c)) + # y-axis for QQ plot
  # first layer: QQ reference line
  geom_qq_line(color = "cornflowerblue") + # displaying line in blue
  # second layer: QQ plot
  geom_qq() +
  # labeling title
  labs(title = "Temperature QQ")+
  # changing theme
  theme_bw()

# creating obhect for elongation QQ plot
elongQQ <-
# baselayer: ggplot
ggplot(data = kelp, # using kelp data frame
       aes(sample = kelp_elong)) + # y-axis for QQ plot
  # first layer: QQ reference line
  geom_qq_line(color = "red") + # displaying line in red
  # second layer: QQ plot
  geom_qq() +
  # labeling title
  labs(title = "Kelp elongation QQ") +
  # changing theme
  theme_bw()

# using patchwork package to display plots side-by-side
tempQQ | elongQQ
```



To figure out which test was accurate to run in this scenario I first checked the linearity of the initial plot, there appears to be a negative linear relationship between temperature and giant kelp elongation rate. Then, to check the normality (I am assuming this data is independent here and we have established that they both are continuous) I created QQ plots of each variable. The QQ plots suggest both variables (and therefore all the data) are normally distributed (centering around the reference line with only slight deviations at the tails) and therefore suggests that a Pearson's correlation test is the appropriate test to perform.

Running the test

```
# running correlation test
cor.test(kelp$temp_c, kelp$kelp_elong,
         method = "pearson") # pearson's test
```

```
	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
```

```
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

d.

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a Pearson's correlation test. I found a strong negative relationship between temperature and elongation rate (Pearson's r = -0.69, t(30) = -5.2, $p < 0.001$, $\alpha = 0.05$)

e.

Through a Pearson's correlation test I was able to determine that there is a clear negative relationship between ocean temperature and giant kelp elongation rate in the sample area. Essentially, this means that within the sample zone in areas with a higher sea temperature we see a distinctly lower elongation date for giant kelp fronds. This is also true in the reverse, meaning that in the areas sampled with a lower temperature the elongation rates of the kelp was seen to be higher.

f.

```r
# running correlation test
cor.test(kelp$temp_c, kelp$kelp_elong,
         method = "spearman", exact = FALSE) # spearman's rank test
```

```
    Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

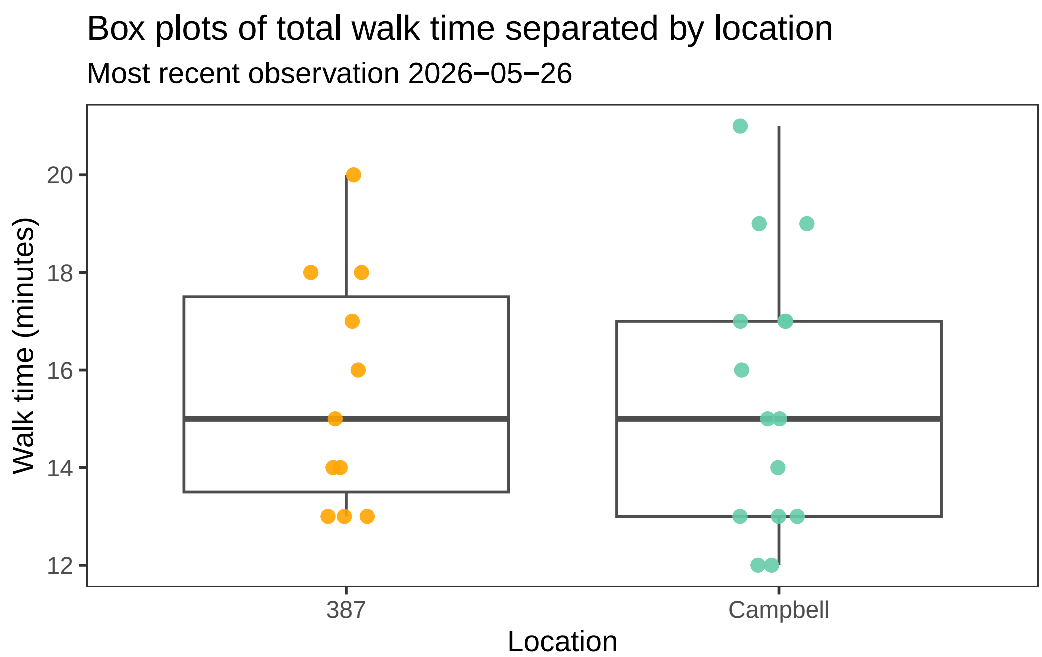
If I had used a Spearman's rank correlation test initially, as opposed to the Pearson's correlation I had chosen, I would have been led to the same decision in regards to the relationship of the two variables and I would have interpreted the results the same. This makes sense as a Spearman test has the same requirements for use as the Pearson but the Pearson is a parametric test, essentially, because we can use a Pearson's test we can also use a Spearman but the Pearson's will provide more information in the test that can be useful.

Problem 2. Personal data

a.

Categorical vs Response

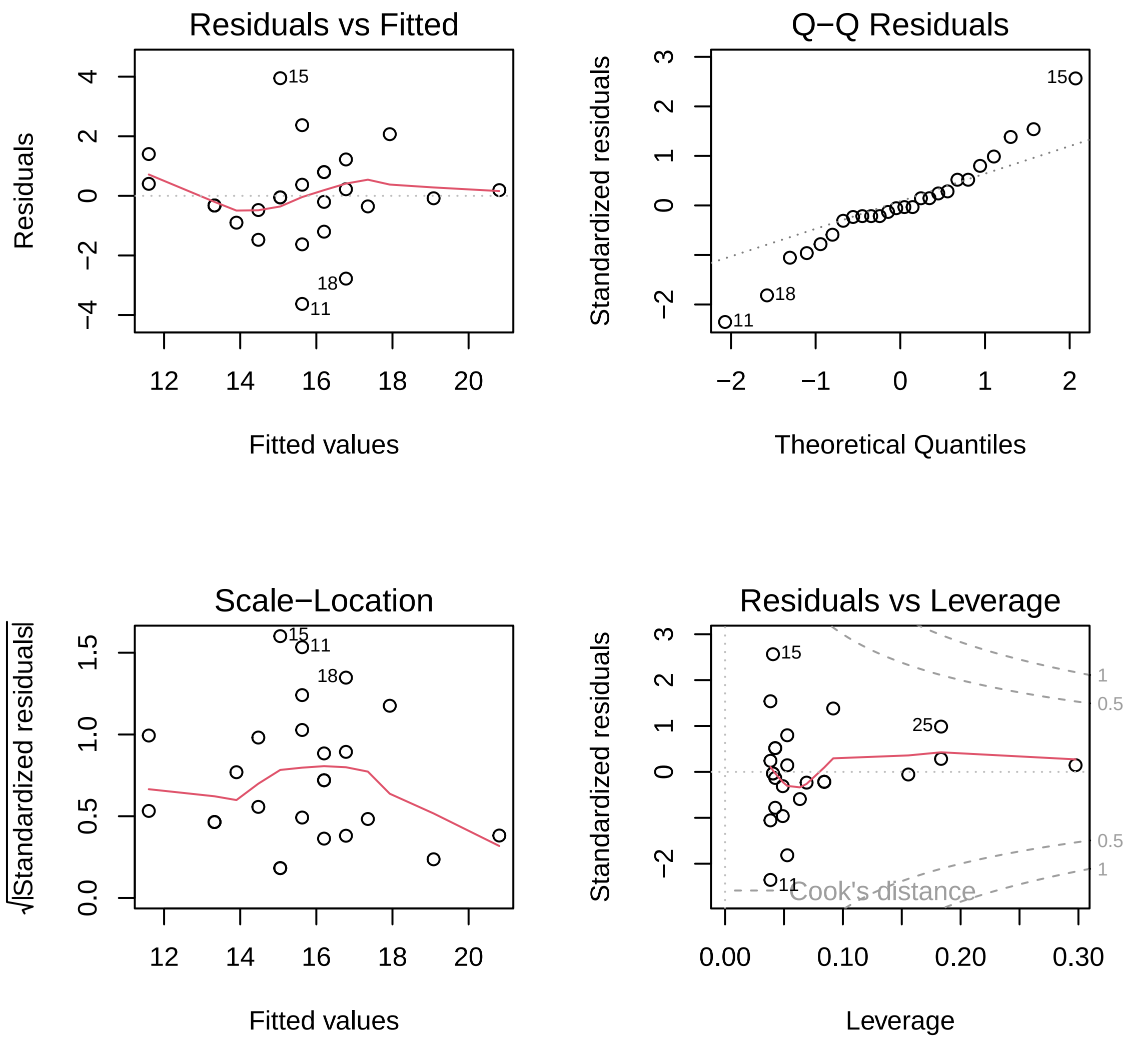
```
# base layer: gg plot
ggplot(data = personal_data, # pulling from personal data frame
       aes(x = location, # x-axis
           y = walk_minutes, # y-axis
           color = location)) + # coloring by location
  # first layer: box plot
  geom_boxplot(color = "grey30", # changing box plot outline color
               linewidth = 0.5,) + # making border lines thicker
  # second layer: jitter plot
  geom_jitter(height = 0, # removing vertical jitter
              width = 0.1, # slight horizontal jitter
              alpha = 0.9, # making points slightly opaque
              size = 2) + # making point size larger
  # changing theme
  theme_bw() +
  # removing legend
  theme(legend.position = "none",
        panel.grid = element_blank()) +
  # labeling x- & y-axes and adding title
  labs(x = "Location",
       y = "Walk time (minutes)",
       title = "Box plots of total walk time separated by location",
       subtitle = "Most recent observation 2026-05-26") +
  # changing colors based on location
  scale_color_manual(values = c("Campbell" = "aquamarine3",
                                "387" = "orange"))
```



Predictor vs Reponse

Model Fitting

```
personalmodel <- lm(
  walk_minutes ~ departure_time,
  data = personal_num
)
par(mfrow = c(2,2))
plot(personalmodel)
```



- residuals normally distributed (QQ)
- residuals are homoscedastic (Residuals vs fitted/scale-location)
- no outliers influencing model coeff estimates (leverage)

```
summary(personalmodel)
```

```
Call:
lm(formula = walk_minutes ~ departure_time, data = personal_num)
```

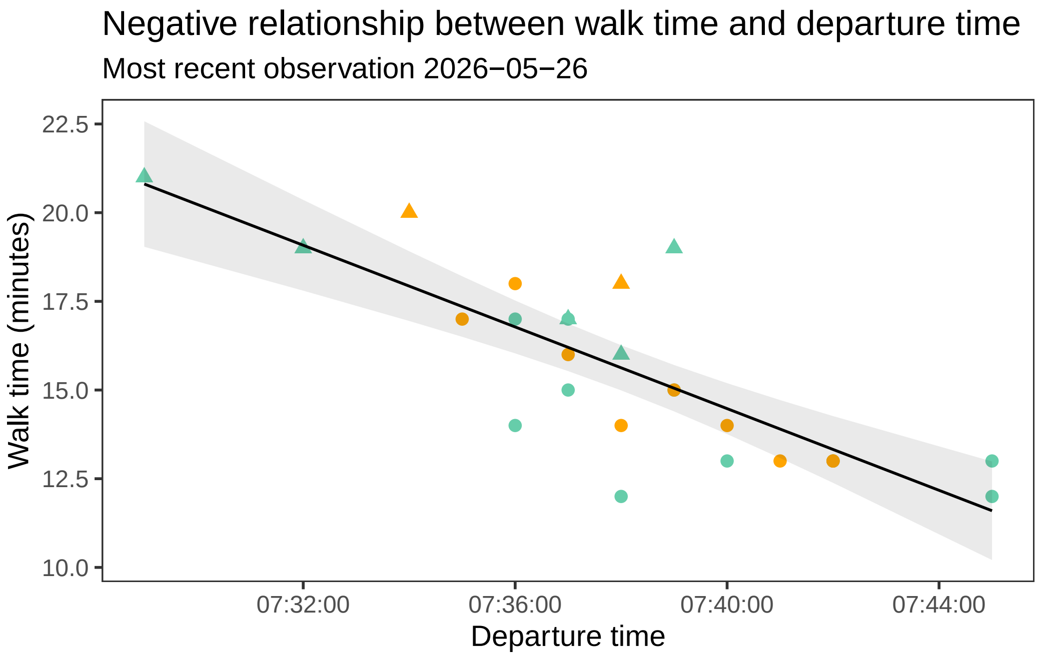
```
Residuals:
    Min      1Q  Median      3Q     Max 
-3.6270 -0.4453 -0.0661  0.6986  3.9486 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)    279.271170  40.049689   6.973 3.28e-07 ***
departure_time  -0.009594   0.001457  -6.585 8.23e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.572 on 24 degrees of freedom
Multiple R-squared:  0.6437,    Adjusted R-squared:  0.6289 
F-statistic: 43.37 on 1 and 24 DF,  p-value: 8.229e-07
```

```r
# creating new object for predictions
preds <- ggpredict(
  personalmodel, # model object
  terms = "departure_time" # predictor
)

# displaying prediction
preds
```

```
# Predicted values of walk_minutes

departure_time | Predicted |       95% CI
-----------------------------------------
         26940 |     20.81 | 19.04, 22.58
         27120 |     19.08 | 17.80, 20.36
         27300 |     17.35 | 16.50, 18.21
         27360 |     16.78 | 16.03, 17.52
         27420 |     16.20 | 15.53, 16.87
         27540 |     15.05 | 14.40, 15.71
         27600 |     14.48 | 13.76, 15.19
         27900 |     11.60 | 10.21, 12.99
```

Figure

```r
# base layer: ggplot
ggplot(data = personal_data,
       aes(x = departure_time, # x-axis
           y = walk_minutes, # y-axis
           color = location, # coloring by location
           shape = stop)) + # changing shape based on stops
  # first layer: points
  geom_point(size = 2) + # making points larger
  # second layer: ribbon representing confidence interval
  geom_ribbon(data = preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              alpha = 0.1,
              inherit.aes = FALSE) +
  # third layer: line representing model predictions
  geom_line(data = preds,
            aes(x = x,
                y = predicted),
            inherit.aes = FALSE) +
  # changing theme
  theme_bw() +
  # removing legend
  theme(legend.position = "none",
        panel.grid = element_blank()) +
  # labeling x- & y-axes and adding title
  labs(x = "Departure time",
       y = "Walk time (minutes)",
       title = "Negative relationship between walk time and departure time",
       subtitle = "Most recent observation 2026-05-26") +
  # changing colors based on location
  scale_color_manual(values = c("Campbell" = "aquamarine3",
                                "387" = "orange"))
```



b.

Figure 1. Median walk time is identical across locations. Points represent individual daily walk times to two different locations, Building 387 (n = 11) and Campbell Hall (n = 15), on UCSB's campus from Isla Vista. Boxplots display median walk time value for each location along with interquartile range and total spread. Location groupings are seperated by color (387: orange, Campbell: aquamarine). Data collected by Luke Simon between the dates of 2026-04-20 and 2026-05-26.

Figure 2. Negative relationship between departure time and walk time. Points represent individual daily walk times to two different locations, Building 387 (n = 11) and Campbell Hall (n = 15), on UCSB's campus from Isla Vista. Point shape represents if a stop was taken on the walk (triangle = TRUE, circle = FALSE), while point color represents location (387: orange, Campbell: aquamarine). Line represents linear model of departure time vs walk time. Ribbon represents 95% confidence interval from predictions. Data collected by Luke Simon between the dates of 2026-04-20 and 2026-05-26.

Problem 3.

a.

For my affective data visualization I want to create a network map of all of my data as nodes. These nodes would vary in shape and color which would each correspond to some detail from the data set. My current thought process, though this will likely change, is that color would operate on a gradient for walk time and node shape would correspond to arrival time (number of sides = minutes after 7:50). I would also include some signifier for if I took a stop and connect each node by location. Node location relative to each other is based on temperature (x-axis) and wake up time (y-axis) and node border color signifies location.

b.

c.

- This draft only includes the first 12 observations